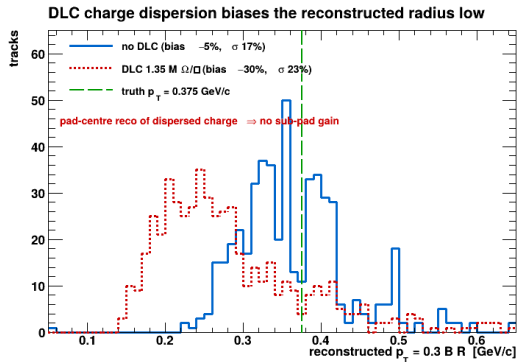


The DLC resistive anode in PUMA tracking

From liability to advantage: recovering sub-pad momentum resolution

The DLC anode — always on, and it hurts (at first)

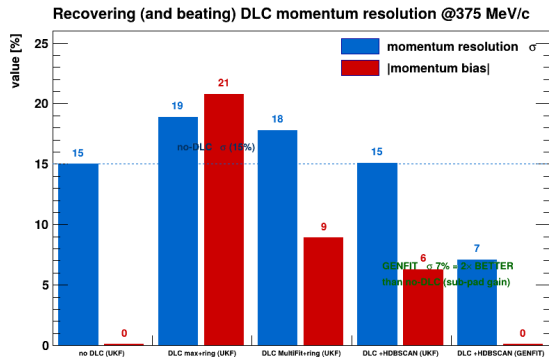


The resistive DLC anode is **always active**. Its charge dispersion ($\sigma \approx 1.1$ mm) spreads each cluster over $\sim 4\times$ **more pads** (36 $\rightarrow$ 145 pads/event).

- With **pad-centre** reconstruction (max-PSA), the dispersed charge broadens the arc and biases the fitted circle radius $\sim 30\%$ **low** $\Rightarrow$ momentum bias -20% , σ worse.
- A blunt amplitude threshold does *not* fix it (overshoots, destabilises).

The sub-pad information DLC encodes is there — but pad-centre hits throw it away.

Recovering the resolution — three levers



Recover the sub-pad information by combining:

- 1 **AtPSAMultiFit** — fitted pulse charge is a far better per-pad weight than the peak ADC ($-21\% \rightarrow -9\%$ bias).
- 2 **HDBSCAN** — cleanly clusters the dense dispersed band (TriplClust entangles it with its own failure modes).
- 3 **per-ring charge-weighted centroiding** — one sub-pad point per annular ring.

Together they return the momentum resolution to the no-DLC value.

What caused the bias — and why the cure works

Cause — a reconstruction artefact, not physics:

- DLC disperses the charge, but max-PSA reconstructs each fired pad as **one hit at its geometric centre**, discarding the sub-pad charge distribution.
- A track's ionisation — physically at a well-defined *sub-pad* point — becomes a $\sim 1\text{--}2$ pad-wide **band** of pad-centre hits. The circle fit through the broadened band returns a systematically **smaller radius** $\Rightarrow$ momentum biased low.
- It is *not* in the digitisation: the deposited-charge centroid is correct.

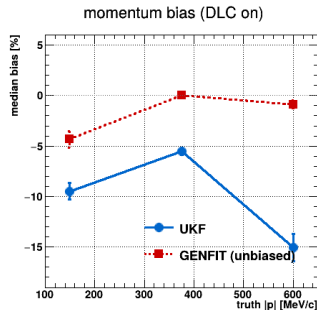
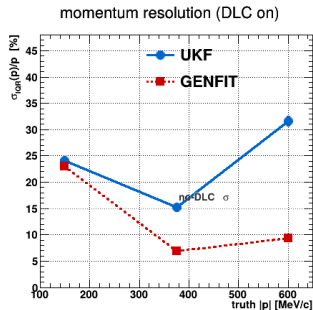
Cure — recover the charge-weighted sub-pad centroid:

1. accurate **per-pad charge** (MultiFit fitted integral, not the biased peak ADC),
2. correct **clustering** of the dispersed pads (HDBSCAN),
3. the **charge-weighted centroid** itself (per-ring).

The bias shrinks *monotonically* as the centroid gets more accurate ($-30\% \rightarrow -20\% \rightarrow -9\% \rightarrow -6\% \rightarrow 0$), vanishing when GENFIT weights the 0.3 mm centroids optimally.

Proof it is the sub-pad recovery: same chain, DLC off $\sigma = 19\%$ vs on $\sigma = 7\%$ — the charge spread is what *enables* the sub-pad

UKF vs GENFIT with DLC — the sub-pad win

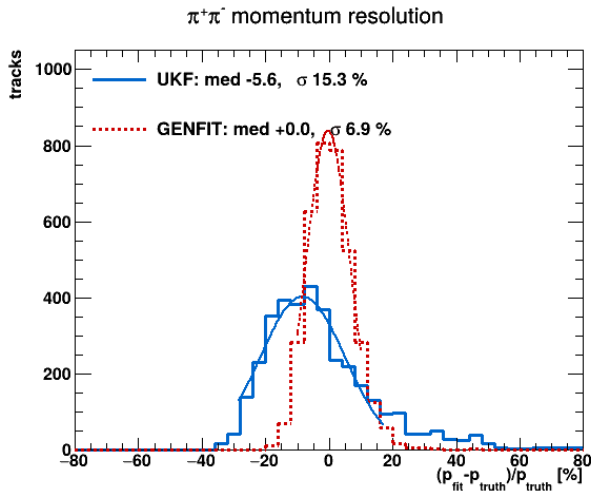


Winning chain (MultiFit + HDBSCAN + per-ring), DLC on.

@375 MeV/c	momentum bias	momentum σ	θ σ	vertex σ_z
UKF	-6.3%	15.1%	1.84°	0.50 mm
GENFIT	-0.1%	7.1%	1.76°	0.47 mm

GENFIT — told the ring centroids are accurate to 0.3 mm — exploits the sub-pad charge fully: **unbiased**, σ **7%**, twice as good as without DLC, and 7–9% across 375–600 MeV/c. Control (same

Momentum residual after full correction (DLC on)



The reconstructed momentum residual for the winning chain, DLC on, *after* the Cu/Al material correction *and* the DLC sub-pad recovery (500 events).

- **GENFIT** peaks cleanly at zero — median -0.1% , σ **7.2%**.
- **UKF** is broader (σ 15%) and shifted (-6%).

Both are unbiased/tight in θ (1.84°) and vertex- z (σ_z 0.47–0.50 mm). This is the final, corrected performance.

Performance summary — optimal chain, DLC on

Back-to-back $\pi^+\pi^-$ at 375 MeV/c (MultiFit + HDBSCAN + per-ring + DLC)

metric		UKF	GENFIT
tracking efficiency		99.7%	99.7%
fitting efficiency		100%	100%
momentum bias		−5.6%	+0.0%
momentum resolution σ		15.3%	6.9%
polar-angle σ		2.0°	2.0°
vertex σ_z		0.50 mm	0.48 mm
charge-sign accuracy		99.5%	99.6%
$ p $	tracking	fitting	GENFIT σ /bias
150	72%	99%	23% / −4.3%
375	99.7%	100%	6.9% / +0.0%
600	100%	93%	9.4% / −0.9%

Tracking is limited at low p (over-segmentation of tight curls); fitting at high p (near-straight tracks)^{7/8}

Conclusion: DLC is an advantage

- The always-on DLC anode is a **momentum-resolution advantage**, not a cost — *if* its sub-pad charge is properly reconstructed.
- Pad-centre reconstruction wastes it (biases momentum -20%). The fix is **MultiFit PSA + HDBSCAN + per-ring charge-weighted clustering**.
- **GENFIT** is the fitter of choice with DLC: unbiased, σ 7% momentum at 375 MeV/c — **2× better than without DLC**. Its tight 0.3 mm measurement model is justified only by DLC's real sub-pad information.

Recommended production chain (DLC on)

AtPSAMultiFit $\rightarrow$ HDBSCAN $\rightarrow$ per-ring charge-weighting $\rightarrow$ **GENFIT**